

逻辑设计大作业

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- 1 写出由一般的“与或式”得到最简“与或式”和“或与式”，以及由一般“或与式”得到最简“与或式”和“或与式”的方法。
- 2 设计一个输血-受血判别组合逻辑电路，当输血者和受血者的血型符合下列规则时，配型成功，受血者可接受输血者提供的血液。
 - (1) A 型血可以输给 A 型或 AB 型血的人；
 - (2) B 型血可以输给 B 型或 AB 型血的人；
 - (3) AB 型血只能输给 AB 型血的人；
 - (4) O 型血可以输给 A、B、AB 或 O 型血的人

由题目得知一共有四种血型：A, B, AB, O, 因此可以用两位二进制数来表示输血者和受血者的血型。下面给出表示形式。

表 1: 血型表示

O	A	B	AB
00	01	10	11

我们用一个四位二进制数 $X_1X_2Y_1Y_2$ 来表示电路的输入, 其中 X_1X_2 表示输血者血型, Y_1Y_2 表示受血者血型。电路输出用一个二进制数 V 表示, $V=1$ 表示配型成功, $V=0$ 表示配型失败。下面给出需设计组合逻辑电路的部分真值表 (当 $V=1$ 时)。

表 2: 需设计逻辑电路的部分真值表

X_1	X_2	Y_1	Y_2	V
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	1
0	1	0	1	1
0	1	1	1	1
1	0	1	0	1
1	0	1	1	1
1	1	1	1	1

由表 2 可列出逻辑函数为“最小项之和”表达式：

$$F(X_1, X_2, Y_1, Y_2) = \sum m(0, 1, 2, 3, 5, 7, 10, 11, 15)$$

由卡诺图化简得：

$$F(X_1, X_2, Y_1, Y_2) = \overline{X_1} \cdot \overline{X_2} + Y_1 \cdot Y_2 + \overline{X_1} \cdot Y_2 + \overline{X_2} \cdot Y_1$$

给出逻辑电路图：

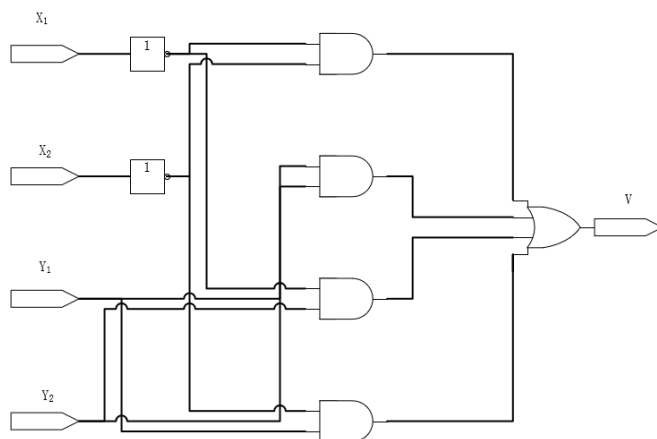


图 1: 电路图

3 设计一个篮球机控制器同步时序逻辑电路

A company asks you to design a controller for a basketball machine that will indicate that a player has won a game if they get a total of 3 points. There are two sensors in this machine. One in the backboard and on the inside of the orange rim. If a player shoots a shot and the ball hits the backboard and the ball goes through the rim, then the play will gain 1 point. If the player shoots the ball and it doesn't hit the backboard and the ball just goes through the rim, then the play will gain 2 points. If the player hits the backboard and the ball does not go through the rim, then the player will lose 1 point. The total score can only go down to zero and cannot become negative. If the ball does not go through the rim or hit backboard, then it will count as a no shot and nothing will happen. Once the player scores three points, the game ends, and a light will indicate that the player has won.

Part A: Make a state table that describes each state in plain English and describe each binary value used.

Part B: Make a state diagram.

Part C: Make a k-map.

Part D: Make the synchronous machine schematic.

You can use either a Moore or Mealy machine and either a J-K Flip Flop or a D Flip Flop.

I was thinking about using 2-bit binary but then I got stuck so I switched to 3-bit binary and what I have looks good to me but now making the k-map is making me doubt what I have so here I am asking for help.

We can know that there are four states through the title, which can be indicated as a two-digit binary number y_1y_2 . Also we can use a two-digit binary x_1x_2 number to indicate the scoring(input). Among them, x_1 represents the state of the sensor in the backboard, and x_2 represents the sensor on the inside of the orange rim. Here is the description.

表 3: Description of y_1y_2

State	y_2	y_1	description
S_0	0	0	Zero point
S_1	0	1	One point
S_2	1	0	Two points
S_3	1	1	Three points

表 4: Description of x_1x_2

x_1	x_2	description
0	0	Score zero point
0	1	Score two points
1	0	Minus one point
1	1	Score one point

Then we can get the state table via the logic of the title.

Current state	Secondary state				Output	Description
	$x_1x_2 = 00$	$x_1x_2 = 01$	$x_1x_2 = 10$	$x_1x_2 = 11$		
S_0	S_0	S_2	S_0	S_1	0	The player scored zero points and the lights went out.
S_1	S_1	S_3	S_0	S_2	0	The player scored one point and the lights went out.
S_2	S_2	S_3	S_1	S_3	0	The player scored two points and the lights went out.
S_3	d	d	d	d	1	The player scores three points, he wins, and the lights come on.

And then we can draw the state diagram.

We chose the D Flip Flop to store our state, so we can make the Activation-Function k-map and the Output-Function k-map.

In the end we can get the Activation-Function D_1 , D_2 and the Output-Function Z :

$$D_1 = x_1 \cdot \bar{y}_1 + x_2 \cdot y_2 + \bar{x}_1 \cdot y_1 = x_2 \cdot y_2 + x_1 \oplus y_1$$

$$D_2 = \bar{x}_1 \cdot x_2 + \bar{x}_1 \cdot y_2 + x_2 \cdot y_2 + x_2 \cdot y_1 = \bar{x}_1(x_2 + y_2) + x_2 \cdot y_1$$

$$Z = y_1 \cdot y_2$$

The synchronous machine schematic:

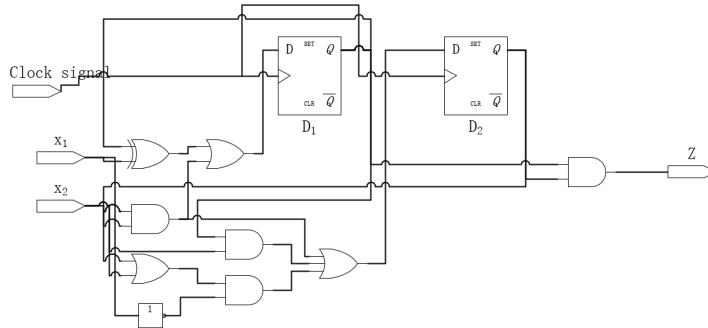


图 2: 电路图

4 使用 74138 (3 线-8 线译码器) 或 74151 (8 路选择器) 或 PROM 或 PLA 设计一个输血-受血判别电路, 当输血者和受血者的血型符合下列规则时, 配型成功, 受血者可接受输血者提供的血液.

- (1) A 型血可以输给 A 型或 AB 型血的人;
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